

ActInf GuestStream #005.1 ~ Martin Biehl

GuestStream | Martin Biehl | 1:49:41

great

hello and welcome everyone to actin flab

today is april 28 2021

it's actin flab guest stream number 5.1

with martin beale

martin thanks so much for joining us on

today's stream

and we're really looking forward to

hearing about your perspective and what

you have to share here

so uh if anyone in the live chat would

like to

ask questions we'll be relaying those to

you during and at the end of the stream

other than that please take it away and

thanks again for coming on

uh thank you daniel um

welcome to my talk everybody uh i will

talk

about uh

or i will talk about at least in the

background

about the paper i wrote with felix

pollock and

rio techni which is called technical

critique of some parts of the free

energy principle

but i will not go directly through this

paper

i will kind of try to give a more

updated perspective

um yeah what else to say i work at uh

aurier in tokyo

and this work is also funded by

temperature world charity foundation

and fedex was supported by the monash university

okay so here's the

content of this talk first i'll make

some preliminary remarks and then

then talk about my perspective basically

on the free energy principle and how

it can maybe maybe split into two parts

uh the construction of agents and the

identification of agents and

then talk about what

the fep agent construction is and what

the fvp

agent identification method kind of is

and then i'll talk about the relation

between those or at least the claimed

relation between those

and then i'll go through some issues

with this agenda identification method

and also with this relation

okay um so yeah like i said the main

publication

uh is

same name as the has the same title as

the stock and

um came out this year

and it resulted from an effort with

uh felix where we we actually set out to

do like a discretized uh version of the

life as we know it paper from 2013

but we failed and then in the end we

wrote this

critique

um so yeah i will i will give this a

in this talk i will give a

my personal view a little bit of the

free energy principle

and why i find it interesting and like i

said i want to go through the paper in detail

um and instead summarize

uh how i currently see

the issues also the issues that we

mentioned in the paper um

but the ffp the free so fvp meets free

energy principle

as while we were writing this paper it

has changed and so

my perspective has also changed

and but uh if you have questions uh

particular questions

uh about the paper then you can of

course ask me

afterwards

okay so here's the

[Music]

maybe a little bit more detailed

overview

so in my opinion the free energy

principle

can be viewed as two parts um

and the first part is basically what

uh maybe you've heard this kind of this

statement that markov blanket

implies uh markov blankets imply

bayesian inference

and i so from my perspective this is

kind of a way to

identify agents in stochastic processes

so this is one part

that is contained in the fep literature

and another part is about how to

construct artificial agents for

for example for reinforcement learning

or partially or observable

markov decision problems and i call this

the constructing agents part
and the claim
the claim is that uh
like if you read papers about this
constructing agent pod
the claim is that they that this
construction is based on
this identifying agent spot so the
construction is based on this
markov blanket's employability and
inference um
part of the literature and i will kind
of argue that this
is has not been established and i also
kind of think it's questionable
hey at least in the current
state then i also present some other
issues with the
fep among those those that i mentioned
in the paper
and these issues so yeah one thing i
want to say is that these issues
they kind of affect would affect i mean
if
if they are if i'm if i'm correct about
those issues then they
affect the usefulness of the free energy
principle
or agent identification so so then we
couldn't
uh identify agents using the fep
and this also would question it as a
theory of what life is or
of life general or life as we know it if
you want
and then um
it would also affect this claim that
uh these three these constructed free

energy

principal agents are somehow based on

first principles

more than let's say some other

reinforcement learning agents

what uh what all of this what all of

these points in my opinion won't

affect is um

that free that these free energy uh

principal

agents like free energy minimizing

agents that are constructed

in the literature that they work or

uh that they work at all basically or

that they can be useful for

some research i mean i think i think

those

i think that these agents work is uh

kind of independent of this whole

markov blanket implies basically

inference part

but maybe i'm wrong but okay

so what's this uh what are these two

things uh

in the construction of agents and the

identification of

agents so like i said i think the

literature can be split into these two

parts

and like how to construct

an agent or a controller

and the other one is how to identify

basically an agent

in a stochastic dynamical system

and yeah so

and like i said also before the the

this first method how to construct um

agents or controller is claimed to

inherit from the second method
and these are still claims that are uh
current so there's a paper that came out
just this year or actually
a week ago so i think it's called
sophisticated inference
okay so what do i mean by construction
of agents or what is this uh what is the
problem of
construction of agents um because that's
a
yeah a different problem than the
identification of agents
that's kind of the first point i want to
make at least two
um problems are different
in general
so yeah in in artificial intelligence
research what
what people are occupied with
uh is the construction of intelligent
agents
for some given environment or task or
like
um and uh yeah so basically what what
these people do in artificial
intelligence
research or engineering is programming
robots
or virtual agents and of course the
robot
is then kind of fixed
a fixed environment or if you have a
virtual world
then this is also a fixed environment
and then if you have some
reinforcement learning problem then you
have kind of a task

okay so um

like i said we in these cases what we

have is

some environment uh that is given and i

uh i write this here as a bayesian

network

and then we construct a controller and

then this what this basically does is we

have some environment states

that i call eater and sensor values

that depend on each other and

we have dynamics of each other so so

basically towards the right the

time increases time step increases um

so we have uh we have some dynamics of

this

environment which tells us how each

changes under the influence of actions

and depending on its previous state

and the environment produces sensor

values and what an agent does

is it takes these sensor values

and updates its uh internal state

which is called μ and produces actions

accordingly so as you see here μ

depends on both the sensories and the

actions

how exactly these edges are

look there's some variability in it but

it's roughly one

one idea like yeah

most of them are roughly equivalent

so like i said or a bit more formally

maybe uh each of t is an environment

state

or an environment or a task is defined

as

by an environment state

sensor values some kind of action
interface
so some dependency on the external
variable so this dependency of the
update of η on the action
and the dynamics which means
if we have a if we are in discrete time
then the dynamics can be specified as
this
markov kernel or conditional probability
of the next or the yeah the next
internal state the next sensor produces
sensor value
conditioned on the pressed internal
state
past sensor value and the action
which is the influence from the patient
and in continuous time we could also
formulate this
and one way to formulate this looks like
this we give the
uh time derivative of the internal state
as some function on its own
uh state and the sensor and the action
plus some noise
and similarly for the
sensor dynamics
and maybe it's not so important but
in general you can have and
an environment can have some kind of
reward
that it gives to the agent as part of
the sensor value or
um it may not
okay and then what's an agent so an
agent has this
uh internal or memory state and it
produces actions

and it also has an interface for
accepting
or sensors that are produced by the
environment
and its dynamics this can also be
written as this kind of conditional
probability
or similarly with time derivatives
in the time continuous state and maybe
the
the main part uh that's going to be
relevant for
some something later in this talk is
that
we can combine the we can combine such
environments and
uh agents
and then they form a markov process so
then we have
uh some joint variable x and the product
of these two
uh conditional probabilities the one for
the environment and the one for the
agent
they induce this um
markov kernel of the joint process
and similarly in the continuous case we
can uh
see the whole thing as one the
combination of the
environment and the agent as all as a
single
uh very as a single
markov process with uh
state x
okay so that's kind of what we
maybe should keep in mind um
some remarks uh on this usually we have

we construct these agents to solve some problems so we have some kind of reward um so this is the case in reinforcement learning for pomdps but we can also have agents that have just intrinsic goals and they don't need any reward from the environment these things are usually called intrinsic motivations and the these the fep agents from the literature they actually kind of have both uh in within them so they have a they contain some part that's like an intrinsic motivation and they also can um they have some part that is similar to a reward and so basically the entrance motivation part is is like a maximization of information gain and the uh instead of reward maximization we minimize some divergence from uh a desired distribution which is good but at least on a high level this can be seen as uh so these two things are kind of combined in the free energy in the expected free energy yeah i mean there are some tricks to this but roughly this is true okay so um so that that's that's what uh i i hope it came across uh what i mean by construction of agents i don't know if there were any questions

uh already
and it's not yet continue thank you
not yet okay so then uh what do i mean
by identification of agents
which is maybe a bit less
uh common practice
so first maybe let's ask the question
what what is an agent
uh in general um
and uh so very roughly
there's a definition by uh
and they say um to be an agent you have
to be
something that's individuated
that can act and make things happen and
can act which just means to make things
happen basically
and uh
it can act to achieve the these actions
are
are performed in order to achieve some
goal so an agent has
also a goal maybe something that has a
goal
an x to achieve actively pursues this
goal
and these these agents that i just
talked about these constructed agents
that we construct in artificial
intelligence
but also in the fvp literature
they are agents because they are kind of
by construction
they are individuated because we kind of
know where their boundary is
uh like if you ask some engineer
what they have to do to construct the
agent they know exactly what to do so

they they seem to know
where the agent uh is or what the agent
is
and of course since we program into them
uh
the intrinsic motivation or some kind of
reward maximization mechanism
they are also co-directed
by construction
um okay so but then
this is not this is just what what
agents are so what's
what's what about identification of
agents
so the the at least for me the
origin of this question is uh the kind
of the problem of agents in physics
so the physicists they tell us that
everything is uh determined
by the laws of physics so that means
everything that happens is made happen
by the laws of
physics and not by any thing
within this
uh within the universe
so in this if they are correct then
nothing within the universe can make
anything happen
but that also means that nothing can act
to achieve a goal because nothing can
make anything happen so
you can't act to achieve some goal
because you can't act in the first place
so this seems to suggest that basically
there are no asians in the universe like
it must be some kind of illusion
at the same time i mean we are here uh
we are agents and we kind of

want to we kind of have goals or at least it seems extremely likely and it also seems like there are things that don't and we built these robots and they seem to be quite real things and um have goals also so it's kind of so maybe there is some way there should be some way to make these two things compatible um so what we want is uh at least if you're if you like the uh physicist's perspective um and also you like you like to think about agents then you maybe you what you want is a formal theory of agents that is compatible with physics and then so one one problem that such any such theory should be able to solve is that if you have some kind of universe and in physics that's basically just a dynamical system uh then what and where are the agents so if you have if you have a formal definition of agents or former theory of agents then this is the question that it should probably be able to answer so here's the universe um and what we want to do is we want to check where inside this long history of the universe

or uh there are these things
that are maybe can be called agents like
this
animal here
a bit more formally so we have some kind
of
in physics we have some kind of
dynamical system we have a state space
and states and we have a dynamical law
that is looked maybe looks like this uh
the time derivative of y is
some function of the so the time
derivative of the state of the universe
is a
function of the current state of the
universe
and then we have
so these things
have trajectories and if we
select one of those trajectories
by some initial condition y_0
then what we would expect from a theory
about a
theory about agents and one that can
also
identify agents then what we would want
is that
this theory gives us like once we give
it such a trajectory of the dynamical
system
or maybe the trajectory and the
dynamical system
that this theory tells us what is the
set of agents
contained in this trajectory and it
should tell us this
including what is basically its body or
its

individual the what is the individual
uh what are the actions and what are the
goals
okay so this is what i
this is kind of the agent identification
problem maybe
and uh the fvp
um does something like this
they start not directly maybe with a
deterministic
dynamical system as the physicists maybe
like but they instead start with
something that's
usually considered as a subsystem of
such a deterministic universe
and this subsystem should obey this kind
of longing equation
um and then the fep
kind of tells us what uh
how to like conditions under which
there's an
agent inside such a thing
so i'm not 100 sure how this relates to
this
to the exact formulation i gave here
where you just get a trajectory and the
system
and it tells you where the agents are
there are
some subtleties to this but but roughly
i think
the free energy principle is definitely
a step towards such a
theory and this part of the free energy
principle
is what i call the free energy principle
identification

[Music]

the free the fep agent identification
method
and it is uh if you're familiar with the
literature
then this is where uh the markov blanket
placed the
fundamental the so-called mark of
nucleation maybe say
okay martin can i ask um
what do we gain or lose by by going to
the lying given
equation what is what is gained or lost
in in that
maneuver relative to the more general
phrasing you had on the previous slide
yeah good question
so i well i i i don't really know
um i all i can say is that
uh
yeah i mean one thing you definitely
gain is that if you
i mean this thing it's kind of uh
hopeless like if you would really
uh if you if we would need the particle
physics
equation
then we would probably never be able to
identify uh
agents because the particle i mean it's
just too complicated and complex
uh to say much about it but
so here this gives us kind of we can
kind of say okay i can ignore a lot of
stuff
and all i have to find is some kind of
part of the system that
uh obeys
some kind of approximate uh equation

basically

and this can be enough um so this is

i guess this is one one advantage

uh

yeah i mean like there's one other thing

i'm not sure

i mean there's one other thing i

sometimes think

uh where this is might be useful for

and that's so what we what we kind of

want

um

is that this agent theory i mean if you

think about it

uh basically a billion year

like a few at some point

there probably weren't any agents inside

the universe

so so at that point

uh so so agents kind of uh

like a full-blown theory of agents they

should

it should allow trajectories

with uh within which agents appear and

basically disappear

or at least they have to be able to

appear

because that's kind of what we think the

universe

happened in the universe and

but this is not um this kind of

appearance and disappearance is actually

not so easy to

to do especially if you would

just kind of naively split up this

system

the whole system into two parts and say

is one part the agent and one the

environment because then you split it
basically

for all time and this is kind of weird
uh because how because you you would
want that the split only works for
a short time or or once they actually
uh is an agent so sometimes i think okay
maybe

maybe these this approximate equation
only works for a short time
and that's an advantage of this also
that

uh there might be short time frames
where this equation works and where they
and others where it doesn't work and
then we would get uh

we could think about whether this means
um the agent appears or not

but actually i'm not so sure what these
d and what the details so

yeah that's all i can really say this i
mean i think

yeah it's uh but it's a very good
question i think

thanks

okay uh any other questions because it
would be

a reasonable point uh not yet but anyone
is welcome to do so

thank you though okay

um so how does

the fep construct agents

i'm not gonna talk about this a lot but

uh just maybe

i'm only going to maybe mention the
things that are relevant

later

and this is basically just the discrete

time free energy minimizing agents
so in the free energy friends in the fep
literature
especially the early ones you also have
um

[Music]

continuous time free energy minimizing
ages but

uh yeah i'm gonna mostly ignore them
here they're not really
there's nothing in particular i want to
say about them but um

so in this discrete time free energy
the in this discrete time fep agents
the agent state μ

consists of hyper parameters
um basically hyper parameters for the
conjugate priors

of the models that this agent is
constructed to

uh use

basically and these hyper parameters
they encode the beliefs or um
both the environment state the basically
a model of the environment
and

a model of the dynamics sometimes
there's no model of the environment
dynamics but

sometimes so that's kind of important i
mean they basically yeah

so the actual state uh consists of these
hyper parameters

um and then what happens

uh what happens with this is that uh in
response to some observation

and past action the agent updates
the beliefs and actually that just means

that it updates these hyper parameters
and this is done via variation of base
and um then the agent basically
uh it also has a belief over actions and
this also gets updated and
this is done in in such a way that it
minimizes the expected free energy
which i think i've mentioned this before
maximizes information gain and kind of
minimizes the divergence from desired
release

um maybe somewhat relevant is that
uh if the model class
that the agent maintains

[Music]

contains the true basically the true
or the truth the true model then
um or an exact model of the
environment and so on then
the belief update this variational
belief update actually is
the same as the exact bayesian update
okay so that's already all i say about

[Music]

the fvp agent construction
and now i'll talk a bit about how the
fep agent identification works
so the the main
relevant literature on this
uh agent identification
are four papers as far as i can see
and the first one is live as we know it
from 2013 and then there's the
monograph uh quite a long
paper i think 180 pages something or 80
page i don't know
lots of pages and that's called the free
energy principle for a particular

physics

then there's a kind of an

extract from that but with small

differences and it's called

markov blankets information geometry and

stochastic thermodynamics

by thomas parr

and

and then there's also a paper

which is written in response to our

criticism that's

called some interesting observations on

the free energy principle

say a question nope

sorry okay so uh

okay so this this last paper was in

response to

our criticism so these i think these

contain basically what the

these contain all the things relevant to

this agent identification problem

um and basically they are the main

papers concerning this mark of naked

thing

there are also some other ones but i

think actually these ones are the

essential ones

okay uh yeah our the paper we wrote

it's mainly concerned with life as we

know it because the second

none of the other papers existed when we

started writing it

uh the this one we energy principle

for particular physics came out while

while we were

maybe yeah while we were writing it

um but yeah as i said i

i will try i will try to

um give a uh my perspective that also includes

uh these newer constructions or approaches or papers

okay

yeah so what is the rough idea of this uh

i can't go through this in detail mainly or among other

re the reasons for this is that i

don't really know i don't really

understand how it works

in detail like i i'm still not

still haven't gone through the whole construction

um because i get usually i get just stuck

before i'm through okay

um the rough idea is so

we have our universe and like i said we find this

uh subsystem that obeys

the language equation

um omega is some kind of markov

and white noise and we also

require that this

subsystem or yeah is such that

not only does it obey this equation but

also that this

equation

has a steady state distribution p^*

okay i think this is just going to be

the same thing again we already saw this

ah okay so now what we're going to do is

we find the subsystem

basically let's say uh we have this

system and it obeys this

um equation i mean so basically

imagine this environment with this
animal
and now we further split up
this into some
oh actually i thought there was going to
be something i'm sorry so we further
split up this system basically we
uh we split it up into
here you you basic i hope you can kind
of see that there's
a central nervous system of this animal
and we kind of split split
up this whole system into relevant
into parts that are aligned with agency
in some way
let's go this way okay okay
now okay
that's what we're gonna do afterwards
in one step but first uh
sorry it's a little bit confusing but
there is uh one
nice fact about
so this should have been before these
images there's one
nice fact about these kind of equations
sorry let me just go back here
about these kind of equations if they
have a steady state distribution
and only if they have a steady state
distribution if they don't then you can
do
the following
so in that case if they have a steady
state distribution then
you can write f of x as this
uh kind of gradient descent
or ascent you
you formulated of the log probability

of the log of the stationary
distribution
and uh yeah so then you also get these
two matrices here in front
uh and r is antisymmetric and γ is
symmetric
um and also yeah you can write this
here the the lower probability the log
uh
a state probability u_m
you can write it or turn it into this
uh negative log probability and that's
the definition of this i you've maybe
seen in the papers
the surprise
okay um yeah yeah
like this is not really trivial this
when exactly you can write this it's not
really trivial um
especially
when this γ
matrix and r matrix don't depend on the
position x which is something that
uh is usually assumed in the fvp
literature
yeah i mean if if the force is linear
or if f yeah
the vector field or sometimes also
called flow is linear then
it works and this also means that around
fixed points of a non-linear
vector field it works okay
can i ask you a question on that last
slide martin yeah um
maybe giving an example of a kind of
system that people
apply it to what is x here and then what
does it mean that r

is anti-symmetric and that gamma is symmetric

okay so

i actually don't know i don't think

anybody applies this to anything

uh let me think

at least yeah i mean like that's

actually one of the criticisms

i have um but we can definitely

apply it to some uh

systems and maybe i'm gonna

look somewhere deep down there oh that

was

i know these pictures they're slow

always get trapped in those right i saw

it okay yeah

here okay so

here's a 2d example of such an equation

and with the gamma and the r so

we say this false let's say this force

is

or this vector field is linear so it's

just a matrix

times the vector

so then exactly then we have this matrix

times the state

and we get this

equation and we can uh

find gamma and r you can compute them

there's a way in the literature

you can also see it in it's also

mentioned in our paper

how to compute those

and you can and we can compute the

stationary distribution

it's some kind of gaussian in this case

and it looks like this

and you see here's the on the left you

see this vector field
and on the right you see the steady
state
distribution here it's called ergodic
density but that's the same thing
and how do these split ups look so
if you look at the
component γ times the
gradient you get this vector field and
if you look at the component
oh this should be oh yeah
 r sometimes this is called q but that's
the same thing
just in case you're wondering if you
look at the
component r then it has this it's
basically the circular
uh component of
this vector field it's hard to see that
this vector field here is
has this circular component but if you
see roughly here you see that they
actually aren't
so this one is actually straight always
cool so the left one is the hill
climbing aspect
and the right side is the is it what's
sometimes called solenoidal
or the iso contours yes so exactly
and um so there is one interesting thing
about this
and that's that uh
this is the so whether
this matrix r is everywhere zero or not
decides whether uh
the steady state is a non-equilibrium
steady state or
an equilibrium steady state

um and this is due to
the i mean so the standard definition of
uh
non-equilibrium steady state and
equilibrium or distinction between
non-equilibrium
steady state and equilibrium state
state is that in the equilibrium
steady state you have detailed balance
and in the non-equilibrium
steady state you don't
and if you have this solenoidal flow
then you don't have you that's
exactly when you don't have
uh it basically destroys the detailed
balance
um yeah detail yeah
so detailed balance means that
basically if you if you make a square
small square here
then the flow uh or two
let's take two neighboring uh
squares and then if it had detailed
balance then
the flow from one square to the other
would always be
the same as the square from the other to
the first from the second to the first
but since here this flow always goes in
in this horizontal direction
or not horizontally in this organization
direction
what goes in from the eye one side is
never compensated
uh in the other direction
so then you can say hey but on the left
side
it also doesn't look like it's

compensated and that
that's that's true but it is actually
compensated by the noise
because this is only uh
this is only the vector field and
without the noise factor so this is only
f of x without the noise factor
and the noise factor in this because
it's gaussian the noise factor is also
it has no curl so
uh this is why
you um so the noise can compensate
this this uh component but it cannot
compensate
all right maybe that was a bit
maybe i shouldn't have taken that much
time to explain this but
it's kind of nice
yeah where are we yeah
okay uh yeah so
this is what uh we get this nice
decomposition and
uh it's kind of needed earlier
we only get it if we have this a steady
state
uh and it plays a
reasonably important role in the
literature okay so
this other thing
that we do with our subsystem is we
split it up
in some way and this is kind of
what the mark of the so-called
markov blanket does
so the but here i'm just going to i'm
not going to talk about what the
marketplace is yet
i'm just going to say what it's uh

so about the work that it's supposed to do

so it will give us um

this split of the state into

four of the state coordinates into four parts

basically a partition of the coordinates and uh we will then call

uh each of the external states as the sensory states

a the active and action states and mu the internal states

sorry someone else just joined was that a question oh no no someone else just joined in

not actually sure but c'est la vie and i'll add a little

uh password just so that no one else joins him

yep go for it

okay uh so so then

what the mark of blanket also does basically it does all kinds of it does a lot of work let's put it this way

because all you need is that you have this subsystem and you have the steady state

at the mark of blanket and then what you get is

that the internal active and sensory states

in this system appear to and

now this is important actively

resist the dispersion of its markov

blanket okay

there's this word markov flanked again

so here you should read boundary

or basic you can also read individuality

that's what
this means in this case um by
engaging in approximate vision inference
so so this
this statement basically has all these
features that we kind of want from an
agent identification
theory because
it tells us that if we have this
under these conditions
and by naming these variables in this
way then
we see that these
i mean it basically it it makes sense to
name these
variables in this way because
they we have um some kind of goal
the goal is to to resist dispersion of
the mark of blankets so basically it's
the goal is kind of persistence
and uh the agent
acts uh to pursue this goal because it
actively resists
this dispersion
um yeah and it is individuated because
we have this uh we know which states
belong to it and which states basically
don't belong to it if this distinction
between external and internal
um yeah and then additionally which is
also kind of nice that this whole
this agent is also doing some kind of
approximate vision inference which
is a bonus i guess
so um this is why i
i say that uh the fep
basically is contains a theory of agent
identification

and um
yeah and this is yeah and this is also
the
this is basically the part that
interests me the most about the
free energy principle
it's yeah okay it's kind of uh
maybe a unique selling point compared to
all other
kinds of uh
to to most other approaches to
constructing art
artificial agents and it's also i mean
there are not many
there are not many agent identification
theories
that i know
and i actually hope that i i know all of
them but
uh yeah actually yeah it's quite uh
these aren't common okay um
yeah i said what this i said this okay
yeah so in summary we just make three
assumptions we say
ah some subsystem has lagerman dynamics
it has a steady state distribution and
it has a markov blanket
and return we have identified an agent
um
yeah nice sounds good
okay so let me uh
is there any question up to now there's
a few more
general ones but feel free to continue
and we'll get them i think at the end
because
this next section will probably be very
helpful

okay so so now
um let's talk about what the relation
between this
fep asian construction and the fep agent
identification
parts of the feb literature
are according to the fvp literature
uh so
the these i kind of said this already
but
uh the thing that makes uh fep
or active inference those are actually
the same thing
as far as i know agent special
is that
not two things or at least two things i
mean
there might be more but i think these
are the main ones
or the most important ones at least
these
fep agents that constructed the
constructed fvp agents
that i talked about before
they can be seen as kind of a plausible
or used
they are used kind of as a plausible
model in neuroscience
and this is related to predictive coding
and the patient brain
but i actually i'm not qualified to talk
about this because i don't know
enough about neuroscience uh but the
other thing that makes them special
is the one of the main points
i mean of this talk is that uh
they are claimed to be based on first
principles

and when this statement that they are
based on first principles
actually means uh refers exactly to this
this uh fap agent identification method
so the first principles are basically
what i just
uh talked about that the
larger equation and the steady state and
the markov
blanket gives you this agent
so let's uh in order to
make sure we're not putting words into
the mouth of the literature
let's read this quote once that's from
the
also sophisticated inference paper that
just came out
the key distinction is that base
adaptive reinforcement learning
considers arbitrary reward functions
while sophisticated active inference
optimizes an expected free energy that
can be motivated from first principles
okay actually here it doesn't say based
but actually in another like you can
look at
this paper and um
in another place it says based on first
principles and also this is not
only this is not only mentioned for the
sophisticated active
insurance agents you will see similar
statements
in the literature in other places
so this is i mean this is kind of
in some way it's super cool if it's true
i think
but i'm not so sure it's true um

okay so some notes

yeah okay so

so this is about

whether actually this how important is

that is this

how important is it for the constructed

agents to be based on these first

principles

so in this also in the sophisticated

inference paper the authors say that

the agents expected free energy

expected free energy

which is the thing they minimize can be

motivated from first place

principles but they also

seem to say that the expected free

energy can also be

i mean yeah actually i mean they say in

in one place they say that it can also

be motivated via information gain

don't and they or

i mean there's there's one part that

this refers to so

as far as i understand it they are

there's a there are two ways to motivate

this expected free energy minimization

and one is

via these first principles and one is

via

some other derivation that involves

information

now i wouldn't say i understand these

derivations

this direction completely but

i do know that there's a paper by cohen

and hotter and they show that

if you that there's a

that that basically some combination of

information gain and reward maximization

uh is you can prove that it's optimal in

some sense

so since this expected free energy

is kind of a combination of those two

things

i kind of think let's put it this way

i'm not surprised that

it has it shows good performance or

something like this because maybe it's

actually so closely related to

the method by cone that it just works

so all i'm saying i guess what i want to

say here is that

in case the motivation via first

principles

fails uh

the fep construction of agents may still

be of interest and might still work

and this also means that all these all

publications

i think there are some but yeah i think

there may be

quite a few all these publications that

uh that are basically based on this

expected free energy minimizing agents

they they need not be without merit just

because the

first principle's motivation failed

so um or in other words

they can still they they can still make

as much sense

basically even if this uh free energy

principle agent identification is wrong

if i could ask a question here martin um

yeah what

what kind of first principles are you

most interested in

and then what kind of a theory of agents
or agency
would be satisfying given your interest
in first principles
yeah i mean i i i think so
so if you
see i'm interested in the first
principles that work
i don't know what the first principles
are but like
maybe one one first principle is that
you can start from physics
like that's like you can you start
with some
well i don't know where five to go up
yeah something like this
you start with a dynamical system and
you then
derive using only
notions that are well defined for
dynamical systems
or within such dynamical systems
um to show
that uh this thing this
system contains agents or not or yeah so
i guess the first principles for
the first principles here are that
okay all we have to believe is
that there are such subsystems that
obey the larger equation and
i guess since the larger equation has
been quite
uh popular in physics
um it has this has this
this this has some strong support um
and maybe that's sufficient to be your
first principal
uh also having steady state

is fine it's well defined there's no i
mean that's a mathematically
well-defined thing we can
in principle check this
so yeah then
the other thing that's used in the
literature is the markov blanket
and i think that's also yeah it's a
it could be a well-defined thing um
i think it's a bit of a like
what it means in the fvp literature
is kind of not it's not really clear and
it changes
and maybe has to change again but
at least the original so if yeah it
depends a bit um
yeah i guess i have to i would have to
show a bit more about the
um
markov blanket but yeah i would say
i yeah for first principles for me means
probably just that you have to be able
to define it for
uh dynamic assist it has to be like all
the notions you use you can only use
notions that are defined
for dynamical systems totally makes
sense it's like if you're going to play
the game of dynamical systems analysis
you can only use those
rules exactly
you can only yeah exactly you have to
you have to construct everything from
there
now you don't get anything else that's
it you get the dynamical system
uh and that's it yeah
okay uh

oh suspense boss
yep that's where i was now
yep i said this need not be there
yep okay so now oh
okay so that's uh a big part of the talk
that's kind of that was kind of my
personal
my current personal view on what's
yeah and how i yeah how i look at the
fvp literature
and what yeah how i split it up like i
said and
uh yeah so now
i would go and talk about
basically the issues i see with this uh
fep
agent identification method but if
there's any questions now
might be what's a good point
that's it for now um definitely after
this we'll see if anyone has any
questions but feel free to type them in
the live chat if anyone has it
okay so the first issue i have is that
this markov blanket assumption is
known to be insufficient
uh by now it's known to be insufficient
maybe in the
maybe at the for the very first paper it
wasn't known to be
insufficient or actually yeah for the
first two papers maybe
no yeah i don't know
yeah yeah i can't really say because it
wasn't me
but yeah you know it's it's it's fair to
say that uh
the first two papers they might not have

known

this okay uh

yeah so so so i'm gonna kind of go

through the

markov blanket evolution a little bit

not not not in any detail but

the mark of blanket evolution in the fap

literature

um maybe i should say

like originally i did i think i don't

have this here

uh i mentioned it down there okay

so originally the the turmark of blanket

comes from judea pearl

from this 1988 book

yeah so it's a it's a technical

uh term

that had a definition before it was used

by the fap literature

okay so and in this in this original

paper the life as we know it

there is actually no definition of like

there's no no

it's nowhere clear it's it's not clearly

said what

is meant what kind of markov blanket is

meant

i mean there's some kind of graph

but markov blankets are

come from directed acyclic grass and the

graph is cyclic so it's actually not

clear what is meant i mean there is

but it's clearly stated that the markov

blanket

should imply this equation

that the vector field

when we split it up into the four

coordinates

also actually so it kind of implies also
that there are these four
we can split up the the coordinate into
four
sets of all the coordinates into four
sets of corners
and um the vector field then
has can be uh has
a certain shape form
namely that the internal
and sensory states do not um
the the vector the vector field
components
of the internal and sensor sensory
states do not
directly depend on the internal states
and the vector field components of the
action and x
action active and internal states do not
depend directly on the external states
maybe i said something wrong but yeah
and we call this the so we we
we call this the vector field condition
or
or yeah i mean i call this the vector
field condition because it's a condition
on the vector field
and in our paper this is condition one
okay so this is kind of i think in life
as we know it this is kind of
this is the only uh real
play is where uh
the markov blanket plays a role and by
the way this is not a mark of blanket
i mean not a technically this isn't a
microphone i mean this isn't
a pearlian markov blanket's condition
um then

in uh at least

in in thomas past paper in 2020

the markov blanket is said to imply

this condition the

independence of the external states

given the blanket states which are the

sensory and the action states so we get

this factorization of the stationary

or steady state distribution

and that's also considered a condition

of the markov blanket

this thing is also implicit

in the this condition is also implicit

in the free energy

principle for particular physics

but it's not explicitly referred to as

the mark of naked so it's a bit weird

but um

this is actually a markov blanket in the

sense of

judy airport so this condition is a true

mark of

markov link and it has

come to be called not only by me

uh the time synchronous thinkerness

markov blanket condition and in our

paper this is condition two

those two conditions this one and

condition one

they are independent of each other like

you they are systems that obey this one

and not this one and system the systems

that obey this one

and not this one

and those systems are not

complicated they are actually linear

systems so they are these orange time

own background

processes where you can easily compute
this
stationary distribution and it's a
gaussian
so you can look this up in the in the
paper
and i think i think one of the i mean
like this is wired speculation but
um i think that uh
the the authors
or carl fristen because he's the only
author
uh kind of thought that this
equate that this condition implies this
condition i think
it seems to me there are various reasons
but it's kind of like i'm not going to
go into it but yeah this
seems to me the case and maybe you could
think i mean i can like you could think
this i mean
it's not completely out of the
question but yeah
so then um we wrote we wrote this paper
uh after we wrote this paper here in
2019
and complained about some
some uh
thing in the some things in the
um
free energy principle for a
particular physics
paper
namely this thing called the conditional
independence corollary
which is supposed to follow from the
vector field condition i think
um is the i mean i think that's the

claim
but we showed that it doesn't and so
then
the markov blanket or then
there an an additional condition was uh
presented
in this paper called
uh some interesting observations about
the free energy principle or something
this is the response to our paper and
this condition is that the
solenoidal component
matrix r or q has this shape
okay that's why i mean i'm going to call
this the sony needle condition
if i ever need it again in this talk but
okay
yeah since this came out after i mean
this was in response to our paper so we
don't really talk about this
um okay so
this is just some additional condition
uh and then i think so
yeah so you can kind of see all of these
like these conditions are all used in
to derive this statement here that
that this statement here
the fep agent identification method
basically
that you get to identify
that we actually get we can identify an
agent that
actively
acts 2 that that acts to uh persists
basically with the goal of
systems okay so we have this
this condition as well and then in
in par in the thomas past paper

and also in the monograph
if we look closely there is actually an
additional assumption made
um in the free energy principle for
particular physics it's
claimed to be a consequence
of the other conditions but that's not
true um
uh we can show this it's kind of easy uh
and in in the par paper it's it's uh
actually said that it's an assumption
so i've known that this is an additional
assumption
and this assumption is that there's some
function σ
and this function σ relates to
variables
and these two variables are
the most likely
external state both β given the
blanket states s_{na} and the most likely
internal state
given the blanket state s_{na}
according to the stationary distribution
the steady state distribution
and the and so σ is uh
the assumption is that that that there
exists
a function σ that maps one that maps
the most likely internal state to the
most likely external state
um and
that that this function doesn't need to
exist you can see
for example i mean it's it's not in our
paper i think we should have probably
put it in our paper but
we didn't

account example is uh you can use
basically the system we pro we show in
appendix d
and then we show one of those linear
linear systems again and for this
the bold μ is just
one half a
and the both η is one half s
and since s of t is not a function of a
of t
and basically uh so they are kind of
so every combination of s and t can
occur so there is no
functional relation between the two and
that means there also cannot be
cannot be any function σ
um so so
this means we kind of have to
additionally assume σ
and also yeah yeah
it and the thing is in my opinion this
 σ might actually be a much stronger
the strongest of all of these
assumptions
that are made like stronger than any
of the other conditions here these other
three conditions the vector field the
tsnbc and the
traditional and the solenoidal condition
i think it does probably the most work
but since i don't really understand the
whole
picture i can't really self-care for
sure but
[Music]
so yeah we'll just start on that um
 σ is sort of the proposed function
that's literally linking

internal and external states so when
we're talking about agents
and the way that their internal
generative model is being linked to
generative processes in the outside
world that's like the key link that's
what we're all about so i totally
i totally see where you're coming from
exactly
that's a good yeah that's a good way to
say it exactly i mean
yeah i think i will i will say this
later again but
let me just say it here already these
three conditions
they are all trivially satisfied by four
independent stochastic processes
so like it seems
uh kind of
unrealistic let's put it this way that
any of these conditions by themselves or
even their combination
would be sufficient to
give us this kind of active
active resisting to anything it's like
or modeling
of the environment or um
asian inference about the environment
because
four independent stochastic processes
don't seem to i mean like it would be
hard to argue that
any of those is doing anything he's
modeling any of the other ones
so uh yeah exactly this is um
but if you have such a then yeah then
yeah there might be something going on
okay so in summary we have these four

conditions the vector field condition
the t as
mbc the sony recognition and the sigma
function existence
and i would say maybe call
those the fep blanket together but okay
okay here i say this now apart from the
sigma function all the conditions are
satisfied by four independent linear
stochastic processes with gaussian white
noise
yeah so oh probably sigma is the most
significant assumption
and yeah i mean it's maybe a petty point
but
i kind of think that calling the
combination of these conditions markov
blanket
is at least i mean by
by now it is definitely misleading i
mean i think the use of this word
microfling it was
misleading before but i think
uh it should probably be dropped
but anyway i'm gonna call it fep blanket
oh sorry this is
so and one more thing i am not
i i cannot say that these conditions
these four conditions together together
are sufficient for
uh the whole
fep agent identification argument
uh just because i don't really i haven't
really
gotten through the whole thing but
we know um that they
are necessary i mean they are used in
these

uh arguments okay
so that's the first that was the first
issue
how am i doing in time actually it's
like already super long
you can go for as long as you want and
we can
go as long as you want already at one
hour on twitch
you can go as long as you want today
martin and then we can always schedule
another
time to speak but this is really really
helpful and i'm sure that a lot of
people are gonna digest it and have a
lot to talk to you about
okay okay then i'm just gonna i'm just
kind of taking
okay all right next one
uh this point is not in our paper um
i guess it has been in the back of our
heads but i mean yeah i think it's
it's really also it's become much more
uh
i've become much more convinced about it
and yeah so i'm gonna
just mention it uh and i think the
it would yeah i think it's a kind of
challenge to the
fep community maybe
and i kind of there if i yeah
i think they are kind of aware of this
by now also
so the thing is that there is no example
for which
all of the fep blanket conditions the
four i just mentioned the four
conditions i mentioned

have been proven to hold so
we don't know any longer an equation
that

obeys all of these conditions
oh actually i forgot the steady state
maybe it should be but yeah
there's no longer an equation with a
steady state that obeys those

[Music]

so what this means is
we don't even know if there's any system
uh that the fep identification method
applies to
or that this statement that uh
in very short that that the fvp blanket
implies

uh an agent

that this is

[Music]

like we don't know that there's a system
for which this is true
so this also means we cannot we can
never

check like there's no system we could
look at and check if
there is actually something doing
bayesian inference and in what way it's
doing it

and that it's actively resisting
dispersion

so this is kind of uh

it's it's not there yeah i mean like it
would be kind of

cool to like so in some way this says
the whole markov blanket story
or yeah the whole as i call it fvp
agent identification story might not
might be vacuous

like this is what this says and i

i think it's kind of a serious point and

it should be

addressed i think okay so maybe some

notes on this

uh so there's this primary audio soup

example in the original paper and it's

used

i mean the life as we know it and it's

also used in the monograph

but there's no proof that this thing has

a steady state or that this steady state

or base the gsmbc the

time synchronous markov blanket

um and also of course there's no proof

that it

has a sigma function because in life as

we know it there wasn't even any sigma

function

also this uh

like you could say maybe the constructed

fep agents are systems that

[Music]

obey all these conditions but and then i

mean maybe there are even some of them

but this connection is never made

explicitly like it's always just

there's this claim that they are based

on this

uh on these first principles and this

all of this but

um it's never shown for example that the

agent environment combination which

forms

this markov process that it has a steady

state distribution

and it's actually very unlikely

that in all of them it has a steady

state distribution

and actually you can kind of like i will

argue that it's actually

not true definitely not for all of them

and also even if uh

this agent environment pair would have a

steady state distribution

we would still

want to see that it obeys the tsnbc

and also that is kind of question

so i should mention that this this is

like the second point

that checking that the tsnbc holds uh

is an idea that's by nathaniel virgo

and uh yeah there's some arguments about

it but i won't go further into it

um

the so furthermore it is never

checked like we never look at the sigma

function which appears to place a

significant role

but we never look at this signal

function in these constructed

uh fvp agents

um what what is kind of

what i'm pretty sure that it says is

satisfied in a reasonable way is the

vector field condition

okay

yeah and maybe this

i don't really know why i say this here

but

yeah like the ah okay so you could you

could yeah okay so so you could

maybe so since me and felix

we use the launch onscreen openback

processes to show various

uh things about um

the free energy principle and we
we know that it satisfies
quite a few of like it satisfies uh
three of these assumptions
so it satisfies the vector field
condition we can make it so that it
satisfies the tsnbc and this solenoidal
condition
but there is no such system like
so basically i'm like if you want to
there's no such system that has a sigma
function so if you wanted to
search for an einstein openback process
that obeys that is basically an
existence proof then i can tell you
it's impossible so this proof is kind of
unpublished but
uh and it's uh mostly due to fedex but
uh yeah maybe maybe i'm gonna maybe
you're gonna put it on archive
at some point it's really not
complicated
um okay so that's this
point now i want to make also
another point that is not in our paper
and then we're going to slowly get to
the end because the last point isn't
very
substantial okay so
yeah what i want to say is that there
are agents in the fep literature
for which
the
the system that they are contained in so
if we form the environment
if we look at the environment and the
agent in combination
as a markov process what i talked about

in the very beginning
then and and the
the agent is this uh fep
agent
then for some of them it's
straightforward to show that they that
this
combination this combine this resulting
markov process
does not have a steady state
and the first thing to note is that if
you have some kind of some monogamously
increasing counter
it can never so if your state contains a
mono
monotonously increasing counter it means
you can never return to a previous state
and this means you cannot have a state
state
because the states are not recurrent and
that means
there is no steady state
so as i mentioned before
the internal state the internal agent
state of
uh fep agents are hyper parameters
and are updated uh using bayesian
inference
variation of bayesian inference but that
also
includes exact patient inference
and so these hyper parameters of x exact
bayesian inference
they accumulate evidence which means
they
count occurrences of events
and they only count up
and they count every and for every event

they count something
so they are counters that always that
are
monotonously increasing
and for example in the sophisticated
inference paper days if you look at
equation
4.5 there is one there is an explicit
uh equation for the updating
of one of the parameters which counts
the occurrences of transitions from
expected environment states to
observations
and no such counter can be contained
within any markov chain that has a
steady state
so this means that this
agent this particular agent in the
sophisticated inference paper
is not within
such a system that has a steady state
so it's not within a system that
[Music]
a subsystem it cannot exist within a
subsystem
that obeys the conditions of the
fap aged identification okay
so uh this
so be be aware that this statement also
is not published uh but i kind of want
to
put it out there because it would be
boring if i just talk about
that's already published and um
so but they i mean like there's some
caution advice what i think but
if you want to look at this at a very
simple example of this

uh i published a paper last year on this
um and so i'm pretty sure and this
it seems pretty straightforward this
argument
and then one more thing there so in this
paper sophisticated inference also
there is some argument about steady
state
the appendix a2
but that steady state just kind of just
it ignores the internal state of the
agent it only looks at the other
variables so i think and
i think that's actually necessary like
there's just no steady state if you
include the internal state of the age
so maybe be aware of this and uh
yeah the two consequences of this are
that
um fep agents
are not identified by the fep agent
identification
method and the and this also means that
the fep
agents cannot be motivated
by this uh fvp agent identification or
or by first principles because they
actually violate these first principles
or in this case at least this particular
age
okay uh i think that was yeah
that was this point the last point is
kind of
uh a bit nitpicky but i think it's also
kind of important for some people
and that is that this uh identification
method
is changing in the literature and you

shouldn't think

for example that the original life as we
know it

a paper cons contains a method that's
very similar to the neurons

i mean the these constructions of the
most likely internal and external states
they don't appear there is no sigma
function

but in the later versions these things
these constructions play essential roles
so these two they are two completely
different methods in

okay and and yeah one of these things we
did in our paper is we showed that this
original method

in life as we know it is kind of wrong
but oh

wow where did i go this was
end so

okay don't worry we're not gonna look at
all these slides

uh okay here we are

okay so we we kind of looked at this and
we think it's it's wrong

uh at least the way it stated um

and yeah okay so so be aware that

if you look at this method like don't

maybe maybe just

ignore the life as we know in paper

that's maybe what i want to say

and also like you can't tell from the
literature that

this method is uh very different i mean
it's kind of

they never say why they changed it and
it's

kind of still repre like still mentioned

as the basis of
these things so be aware it's kind of
maybe it's a waste of time reading this
paper
okay that's it uh
in summary what i said is that we can
separate the
literature into agent identification
fep agent identification and afep agent
construction
and that the fep agent identification
method
requires significantly more assumptions
than just
whatever the mark of blanket was in the
beginning
uh that this uh agent identification
method misses an example that would
prove
that it that such systems actually exist
such systems with agents that systems
with such agents actually exist
and it requires a steady state
that at least some of the fvp agents
definitely don't satisfy
the method has changed from its original
but also note it could still be correct
i don't know like maybe there are enough
assumptions you can make and maybe they
are still systems that contain such
asians
uh it could all still work out but i
haven't made it through
all of these derivations so they may
also
be like i said there may be even more
necessary
okay and the last point

that fep agent construction the fvp
agent construction is probably
unaffected
by any of these problems of the fep
agent identification method
or the mark of blanket stuff if you want
okay and it can be yeah okay and this
method
can be probably independently motivated
by some other arguments
that are not first principles if you
like okay
that's it i thank a lot of people
for talking to me about this stuff
uh yeah and that's it thank you very
much for your attention
if you made it all the way especially if
you made it all the way
thank you martin maybe you can unshare
and i'll just ask a few
of the questions from the chat in
closing
okay it's is this the
where is the unshare is it the same as
um maybe go back to the jitsi and just
unshare
it yeah cool it's great
great nice cool
okay perfect so great um
thanks again that was really awesome and
i hope people
paid attention so the first question is
from
peter and peter asked the distinction
between
agent identification and agent
construction is very nice
if i wanted to use that distinction in

my own work should i cite your paper
or is this distinction made elsewhere
too
i'm wondering if i
if i mention this anywhere
i don't know actually uh i don't know i
mean like
it's in some way it's just
yeah it's kind of like a
yeah it's kind of like an observation
that makes a lot of sense and it's been
yeah i don't know no i like if i if i
think of something
maybe i don't know can i see you are you
on twitter
i don't know uh maybe i'm gonna post it
on twitter follow me on twitter
i think it's it's kind of an obvious
thing uh and i'm not the only one who
has uh
i'm definitely not the only one who has
uh observed this
i yeah probably there's somebody in the
literature that
did it in 1950 but yeah i don't know
like i'm not going to be mad at it's you
if you don't cite me let's put it this
way
thanks for the response another question
from peter
is how much should the agent
identification
and agent construction depend on each
other
in other words how big is the problem if
they don't and how
important is it that it comes from first
principles

anyways can i say the first sentence
again i'm sorry uh how much
should the agent identification and
agent construction
depend on each other
okay so are those like two separate
lines or one integrated hole
yeah good question i mean
i guess
i mean in i guess ideally
there would be some kind of relation but
uh on the other hand so we want to
construct agents
like so so i in my opinion the
bacteria are agents but when we do
agent construction we don't really care
like we don't have to
follow the principles of bacteria
necessarily
if we don't like either we have i mean
we might be
we might have to because that's just how
physics works or
due to the constraints of physics but if
we don't have to
then nothing should stop us just
constructing any kind of thing that we
want i mean like
like if we uh yeah so i guess they are
not
necessarily dependent i mean
yeah at least i can't i can see this so
on the other hand i guess it would be
it would also be nice if you have some
kind of
yeah if you still have some agent
identification method which
comprises all possible agents like all

agents that you construct
should be uh examples of
your like should be detected or
identifiable
by your agent identification method
so okay now i'm gonna take back what i
said before they should be independent
i hope that i hope this helped a little
bit yep and
and i'll just give one i think there's
probably more to say but that i can't
at the moment i can't really think of it
so here's one final question and is a
question i had but i'll use the phrasing
of
peter here and so this will be our
closing question hopefully
the beginning of something new in the
beginning of moving forward peter wrote
i suppose the mathematics in free energy
principle and active inference could be
taken as
a work in progress are there suggestions
for how to solve or
work with the issues that you're raising
just any thoughts on this could be nice
so where do we go as learners and doers
and
practitioners and communicators in
active inference
so so yeah like i said i mean
on the one hand you can if you if you
construct agents you can just ignore
what
all of this stuff i said because in my
opinion you're not
relying on this agent identification
method anyway

so so there's no problem like of course
you could maybe wonder
what this base optimal that is sometimes
mentioned in the literature
of the fep what this really means
uh and where it comes from like i think
that's
i find it i always think it's a little
bit vague
what this really means and it's actually
not such a trivial question what base
optimize but
or what optimize um but yeah mostly you
can ignore
this fep identification but if you are
interested in the fep identification
then yeah i mean that
i guess the the the thing is
look at uh either
i mean try to understand where it's
actually going
i mean like i still haven't really
understood these later parts of the
um
[Music]
the monograph where
everything is supposed to come together
because i always get stuck
um before in some equations that i don't
understand and uh
then i think they're wrong but i'm not
sure they're wrong yet
as i would have mentioned that but
there's
some way i don't i'm i suspect they're
wrong but
yeah i'm still not sure and
but yeah i guess that's still like i

don't really see like if you want to
save
something of this method then look at
this but
another another other way is maybe
to think about uh
think about this for yourself think
about what it what are the
conditions you think under which
something
is doing asian inference or something
like this
and ignore all the math by then
that's already in the literature and
then try like once you figured it out
for yourself
maybe uh try to come back to the
literature and compare it or something
because
i sometimes think people rely
too much on what this
what the literature says and i'm not
really sure
that actually what's in the literature
is like it doesn't
strike me
as a very straightforward argument
um and i like straightforward arguments
i think straightforward arguments are
usually better
um like maybe there isn't a
straightforward argument but i kind of
think i always think there's a
straightforward argument
um but that's just my opinion yeah it's
off
well it's a nice point that the
literature is a scaffold for our own

thinking and development the literature
is not uh
a road that's been paved with gold and
it's just perfect in the past
it's actually always a work in progress
and always a scaffold
so thanks so much for sharing this
really work in
progress not that it isn't uh with
landmarks that you've published but
it is a work in progress in the bigger
sense so
also thanks for staying up late to bring
us this information
and we hope that in future live streams
we can
unpack this more and hear updates as you
continue to progress on this line of
research
okay yeah thank you very much for having
me thank you martin
peace all right bye peace